
Improving Emotional Text-to-Speech in Low-Resource Languages via Reinforcement Learning

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1 Project Overview

1.1 Profile

Track: 3. Application

Abstract and project goal: We propose a novel reinforcement learning (RL) framework integrated with the IndexTTS2 architecture to enhance the emotional expressiveness of Text-to-Speech (TTS) systems[Zhou et al., 2025]. Traditional TTS models often suffer from catastrophic forgetting of emotional capabilities when fine-tuned on low-resource languages, leading to flat and inexpressive synthesized speech. Our method addresses this degradation by leveraging task-specific RL rewards to awaken and preserve the emotional expressiveness of low-resource speech models. By incorporating Valence-Arousal (VA) modeling and emphasis prediction, our approach ensures fine-grained emotional control. Ultimately, this project aims to deliver a robust, highly expressive TTS system capable of maintaining rich prosody and recovering emotion across diverse linguistic resources.

TL;DR (“Too Long; Didn’t Read”): This project introduces an RL-based framework to recover and enhance fine-grained emotional expressiveness and prosodic control in low-resource TTS models, mitigating the issue of catastrophic forgetting.

1.2 Motivation

The demand for controllable, highly expressive speech generation is rapidly growing across various industries, including professional dubbing, news broadcasting, and gaming. While large-scale models demonstrate strong capabilities, maintaining these capabilities when adapting to new, data-scarce domains remains a critical hurdle.

- **Why is the problem interesting?** Current industries require nuanced, human-like voice synthesis. However, aligning these high-resource models to low-resource languages without losing their inherent emotional expressiveness presents a unique intersection of transfer learning and sequential decision-making in RL.
- **Challenge 1: Emotion Forgetting After Fine-Tuning.** When high-resource speech models are fine-tuned on low-resource datasets, they typically experience catastrophic forgetting, resulting in a significant degradation of their prior emotional generation capabilities.
- **Challenge 2: Inability to Express Prosodic Rhythm.** Existing models struggle to accurately model the natural cadence, pauses, and pacing required for dynamic emotional expression.
- **Challenge 3: Inaccurate Emphasis and VA Prediction from Text.** It is inherently difficult for models to automatically and precisely infer local emphasis (stress positions) and continuous emotional intensity (Valence-Arousal variations) relying solely on textual input.

- **Why the problem remains open or unsolved?** Low-resource languages inherently lack the massive, emotionally annotated datasets required for traditional supervised learning to capture complex emotional distributions. Consequently, models cannot easily generalize emotional prosody to these languages. Relevant prior works attempting to address speech emotion and expressiveness include IndexTTS2 [Zhou et al., 2025], IndexTTS2.5, CosyVoice3, emotion2vec [Ma et al., 2023], and various heterogeneous model ensemble techniques.
- **State-of-the-Art (SOTA) Method:** Currently, IndexTTS2 stands as the state-of-the-art framework in this domain, offering a strong autoregressive foundation for zero-shot synthesis and feature decoupling [Zhou et al., 2025], which we build upon to reawaken emotional capacities via RL.

2 Problem Formulation

We formulate the emotionally expressive and duration-controlled text-to-speech synthesis task as a fully observable Markov Decision Process (MDP). Within the IndexTTS2 framework, the environment is defined by the autoregressive Text-to-Semantic (T2S) generation process, where the agent (the language model policy π_θ) generates a sequence of discrete semantic speech tokens token-by-token. At any step, the state s encompasses the available conditioning information: the input text sequence, a speaker timbre embedding c , a duration constraint embedding p , and a target global emotion embedding e . The action a represents the generated semantic token sequence, which is subsequently converted into a mel-spectrogram by the Semantic-to-Mel (S2M) flow-matching module.

To achieve fine-grained prosodic control without compromising speech naturalness, the RL learner receives a multi-objective scalar reward signal $R(s, a)$ upon generating a complete sequence. This feedback is composed of three task-specific components: an Emotion Classification Reward to ensure categorical accuracy, a Global Emotion Intensity Reward measured by the distance to a neutral centroid in the Valence-Arousal-Dominance (VAD) space, and a Local Emphasis Control Reward calculated from the maximum relative pitch (f_{pitch}) and energy (f_{energy}) of target emphasized words.

The optimization problem seeks to maximize the expected reward using Group Relative Policy Optimization (GRPO). Specifically, the objective function aims to optimize the policy parameters θ by maximizing $\mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta} [R(s, a) \nabla_\theta \log \pi_\theta(a|s)]$, subject to a Kullback-Leibler (KL) divergence constraint $\beta \text{KL}(\pi_\theta(\cdot|s) || p_{\text{SFT}}(\cdot|s))$ to prevent catastrophic deviation from the base Supervised Fine-Tuned (SFT) model. We operate under the assumption that the discrete semantic token space produced by the T2S module is inherently expressive enough to implicitly discover and capture subtle emotional intensities and local emphasis variations prior to acoustic reconstruction.

3 Empirical Evaluation

To rigorously evaluate the effectiveness of our proposed RL-enhanced IndexTTS2 framework in recovering and controlling emotions for low-resource languages, we employ a combination of objective and subjective evaluations.

3.1 Performance Metrics

We plan to adopt several key performance metrics to comprehensively assess speech intelligibility, speaker fidelity, and emotional expressiveness.

- **Objective Metrics:**
 - **Word Error Rate (WER):** Evaluates speech intelligibility and articulation accuracy using ASR models like FunASR or Whisper [Radford et al., 2022].
 - **Speaker Similarity (SS):** Computed as the cosine similarity between speaker embeddings extracted from a pretrained speaker recognition model to ensure the timbre is maintained after RL fine-tuning.
 - **Emotion Similarity (ES) / Accuracy:** Measured using the open-source emotion2vec model to objectively verify if the generated speech matches the target emotion category.
- **Subjective Metrics (MOS Framework):**
 - **Multidimensional MOS:** We will conduct human evaluations based on a 1-5 scale for Quality MOS (QMOS), Similarity MOS (SMOS), Prosody MOS (PMOS), and Emotion MOS (EMOS).

- **Emotion Intensity Test (EIT):** A pairwise comparison task where listeners identify the stronger emotional sample among weak, medium, and strong synthesized speech.
- **Emphasis Accuracy Test (EAT):** Assesses the consistency between the model’s predicted emphasis positions (local prosody) and those perceived by human listeners.

The combination of EIT and EAT is crucial as our RL method explicitly optimizes for fine-grained intensity and local emphasis, which standard MOS cannot fully capture.

3.2 Baseline Methods

We compare our method against several state-of-the-art zero-shot and emotionally controllable TTS models.

- **CosyVoice2** [Du et al., 2024]: A highly scalable, LLM-based zero-shot text-to-speech synthesizer that supports predefined emotion instructions.
- **IndexTTS / IndexTTS2** [Deng et al., 2025, Zhou et al., 2025]: The foundational autoregressive models of our work, providing baseline performance for duration control and semantic generation before RL enhancement.

3.3 Benchmark Tasks or Datasets

To simulate the low-resource fine-tuning scenario and evaluate the catastrophic forgetting of emotions, we utilize the following datasets:

- **Base Pre-training Datasets:** The model is initially pre-trained on large-scale datasets such as the Emilia dataset and LibriSpeech to establish robust zero-shot capabilities.
- **Emotional Fine-tuning Datasets:** We utilize the Emotional Speech Database (ESD) and the Espresso dataset, which provide rich annotations for discrete emotions, continuous intensities, and emphasis.
- **Low-Resource Simulation Task:** We will construct a specific low-resource task by taking a subset (e.g., 1 to 5 hours) of a target low-resource language corpus. The model will be fine-tuned on this subset, followed by our RL optimization stage, to measure its ability to retain and express the emotional traits learned from the ESD and Espresso datasets.
- **Standard Evaluation Sets:** For standard synthesis quality and stability testing, we will report metrics on SeedTTS test-zh, SeedTTS test-en, and LibriSpeech-test-clean.

4 Methodology

Our methodology introduces an RL-based fine-tuning stage integrated into the pre-trained IndexTTS2 architecture. The core idea is to treat the autoregressive generation of speech representations as a sequential decision-making process. By defining task-specific rewards based on emotional expressiveness (Valence-Arousal) and prosody (emphasis), we guide the model to retain and awaken its emotional capabilities when adapting to low-resource languages.

4.1 Reinforcement Learning Formulation

The generation of acoustic tokens is formulated as a Markov Decision Process (MDP):

- **State (S_t):** The hidden states of the IndexTTS2 autoregressive decoder, incorporating the input text, speaker conditions, and previously generated speech tokens up to step t .
- **Action (A_t):** The predicted acoustic token at step t , sampled from the policy distribution $\pi_\theta(A_t|S_t)$.
- **Reward (R_t):** A composite reward signal provided upon the completion of a sequence or segment. The reward comprises two main components:
 1. *Emotion Alignment Reward:* Evaluated using a pre-trained emotion recognition model (e.g., emotion2vec) to measure the alignment between the target Valence-Arousal (VA) state and the generated speech.
 2. *Prosody and Emphasis Reward:* A penalty or bonus based on the correct placement of stress and natural rhythm.

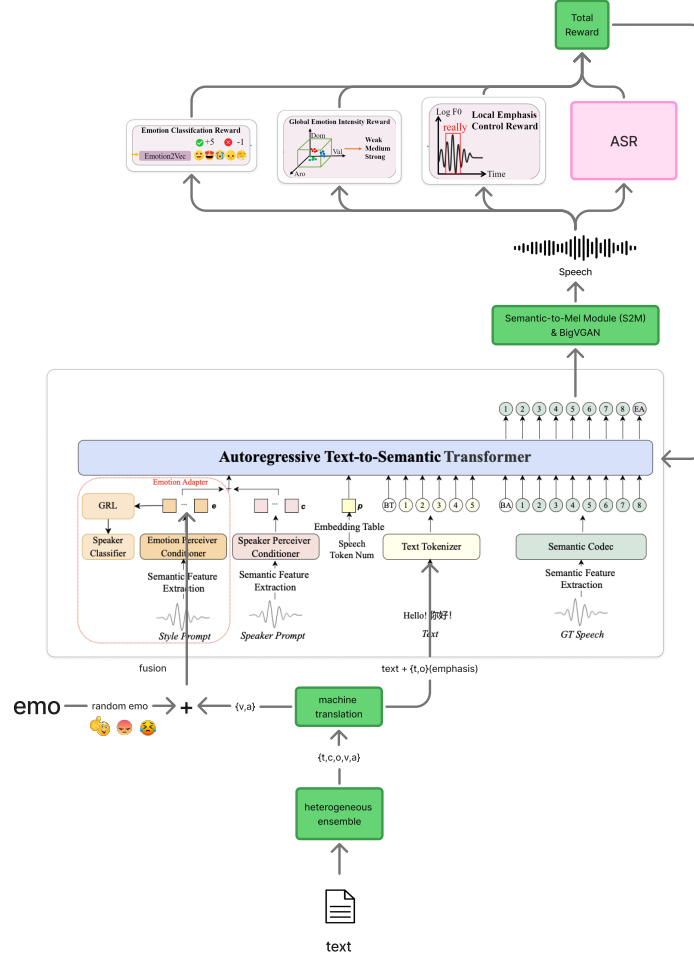


Figure 1: High-level architecture of the proposed RL-enhanced IndexTTS2 framework. The generator outputs acoustic tokens, which are first decoded into speech waveforms before being evaluated by the emotion and emphasis reward models. These rewards are used to update the policy via Group Relative Policy Optimization (GRPO), mitigating catastrophic forgetting.

4.2 Alleviating Catastrophic Forgetting

To address the degradation of emotional capabilities during low-resource fine-tuning, we initialize our policy π_θ with the weights of the high-resource pre-trained IndexTTS2. We employ Group Relative Policy Optimization (GRPO) mixed with a Kullback-Leibler (KL) divergence penalty against the original pre-trained reference model. This ensures that the model explores localized linguistic adaptations while strictly preserving the high-level emotional generation prior.

5 Experimental Results

5.1 Experimental Setup

To assess the synthesized speech, we conducted subjective evaluations using the Mean Opinion Score (MOS) on a scale from 1 to 5 (where 5 represents the best performance). The evaluation panel consisted of 12 members from the SARC laboratory. The test corpus involved 20 sentences for evaluating speech intelligibility and naturalness, and an additional 10 sentences specifically for evaluating voice similarity. In total, the subjective evaluation yielded 600 rating samples ((20 * 2 + 10) test items $\times$ 12 evaluators).

5.2 Evaluation of Baseline Methods

Table 1 shows the subjective evaluation results of the baseline method, CosyVoice2, on Taiwanese TTS. The model achieves a similarity MOS of 4.16, an intelligibility MOS of 4.41, and a naturalness MOS of 4.37, resulting in an overall mean MOS of 4.31. These results indicate that CosyVoice2 performs well in generating high-quality Taiwanese TTS with good similarity, intelligibility, and naturalness.

Table 1: Subjective Evaluation (MOS) on Taiwanese TTS

Model	Similarity MOS $\uparrow$	Intelligibility MOS $\uparrow$	Naturalness MOS $\uparrow$	Mean $\uparrow$
CosyVoice2	4.16	4.41	4.37	4.31

6 Expected Contributions of Each Team Member

- Hao-Chun Hsieh : 34%
- Pao-Hua Hsu : 33%
- Nien-Tse Wu : 33%

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